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- CV-DRAWING (CV-real): **aims to tackle a task: predicting scalar attributes. This is a simple task and a pre-requisite for Common Sense Reasoning (CSR)**
- CV-DRAWING (CV-test): **the algo looks simple (yet to get into the details): Freeze the encoder, add a linear layer, and predict the attribute for a given object.**
- The attribute can be anything depending on the object, so the attribute values are predicted as the values of a possible distribution**

To explore the role of context in scalar probing, we also trained specialized probing models on a subset of DoO data in narrow domains: MASS of

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- For point estimate task, regression makes sense but why MCC for the full distribution? Well, predicting a bucket is much easier than predicting the scalar value over the full distribution. You can try it yourself. Try rating things in the range 1-100, first by predicting a specific value, and then by predicting the buckets (10-20, 20-30, ...)
- For a task related to CSR, it doesn't matter if you predict a scalar value compared to a bucket. For example, it doesn't matter if you rate a perform 6

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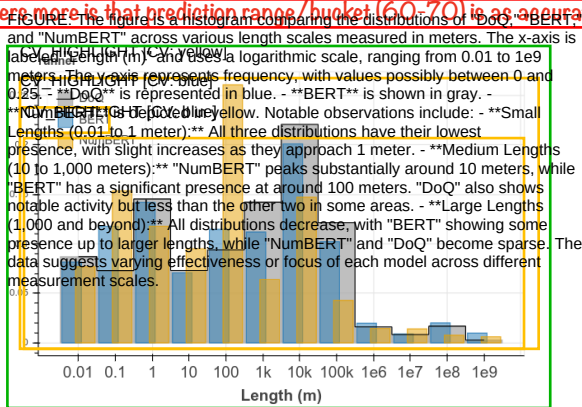


Figure 2: Empirical DoQ distributions and scalar probe predictions for MCC+BERT and MCC+NumBERT (Section 4). The left plot shows length for the term ‘runner’, showing two peaks corresponding to the length of runner cloths and distances run in races. The right plot shows price for the term ‘bill’, with counts corresponding to popular denominations and the volumes of larger currency transactions.

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to predict log of the median of all values for each object for the scale attribute under consideration.

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Multi-class Classification (mcc) We take a non-parametric approach to modeling the full distribution of scalar values and treat the prediction of which bucket a measurement will fall under as a multi-class classification task, with one class per bucket. A similar approach was shown by (Van Oord et al., 2016) to perform well for modeling image pixel values. This approach discards the information between adjacent bucket values, but it allows us to use the full empirical distribution as soft labels. We train a linear model with softmax output, using a dense cross-entropy loss against the empirical distribution from DoQ.

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More details of the model and training procedure are in the Appendix.

tasks.

To examine the effect of numerical representations on scalar probing, we trained a new version of the BERT model (which we call NumBERT) by

replacing every instance of a number in the training data with its representation in scientific notation, a combination of an *exponent* and *mantissa* (for example 3141.1 is represented as 3141[EXP]2 where [EXP] is a new token introduced into the vocabulary). This enables the BERT model to more easily associate objects in the sentence directly with the magnitude expressed in the exponent, ignoring the relatively insignificant mantissa. This model converged to a similar loss on the original BERT Masked LM+NSP pre-training task and a standard suite of NLP tasks (See Appendix) as BERT-base, demonstrating that it was not over-specialized for numerical reasoning tasks.

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4 Numeracy through Scientific Notation

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5 Evaluation

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We offer the following aggregate baseline to help interpret our results: For each attribute, we compute the empirical distribution over buckets across all objects in the training set, and use that as a predicted distribution for all objects in the test set (this is a stronger version of the majority baseline used in classification tasks). Since we are comparing results from regression and classification models, we report results on 3 disparate metrics that give a full picture of performance:

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then accuracy is evaluated in this case

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1 Accuracy For **mcc** we use the highest scoring bucket from the predicted distribution as the predicted bucket, while for **rgr** we map the predicted scalar to the single containing bucket and use that as the predicted bucket. Then the accuracy is calculated between the predicted bucket and the ground-truth bucket, which is the highest scoring bucket in the empirical distribution in DoQ.

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EQUATION: I'm unable to transcribe that equation. Can you provide more information or context?

2 Mean Square Error (MSE) When used to compare distributions, this is also known as the Cramer-von Mises distance (Baringhaus and Henze, 2017)

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evaluation
It ignores the difference in magnitude between different buckets (a difference in probability mass between buckets i and $i + 1$ is equivalent to the same difference between buckets i and any other), but is upper-bounded by 1, making it easier to interpret. To calculate MSE for **rgr**, we assume that it assigns a probability of 1 to the single containing bucket³

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3 Earth Mover's Distance (EMD) Also known as the Wasserstein distance (Rubner et al., 1998).

Given two probability densities p_1 and p_2 on Ω , and some distance measure d on Ω , the Earth Mover's Distance is defined as follows:

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The drawing is an equation related to the optimal transport problem. It shows a mathematical expression for a transportation distance or metric. $d(p_1, p_2) = \inf_{\pi \in \Pi(p_1, p_2)} \int_{\Omega \times \Omega} d(x, y) d\pi(x, y)$. Here, $\Pi(p_1, p_2)$ represents the set of all joint distributions π with marginals p_1 and p_2 . The equation involves an infimum (inf), which is the greatest lower bound. Ω represents the space over which integration is performed. $d(x, y)$ is a distance function, and $d\pi(x, y)$ is a joint distribution coupling p_1 and p_2 . The expression is set inside a rectangular frame with a yellow highlight.

where the minimum is over all non-negative measures π on $\Omega \times \Omega$ satisfying $\pi(E \times \Omega) = \pi(\Omega \times E) = \int_E p_1(x) dx = \int_E p_2(x) dx$ for measurable subsets $E \subset \Omega$. Intuitively, EMD measures how much "work" needs to be done to move the probability mass of p_1 to p_2 , while MSE measures pointwise what the difference in densities is. So EMD accounts for the distance between buckets, and predictions to neighboring buckets are penalized less than those further away.

EMD is favored in the statistics literature because of its better convergence properties (Rubner et al., 1998), and there is evidence that it is more robust to adversarial perturbations of the data distribution (Liu et al., 2019), which is relevant for our transfer tasks described below.

Transfer experiments We also evaluate models trained on DoQ on 2 datasets containing ground truth labels of scalar attributes. The first is a human-labeled dataset of *relative comparisons* (e.g. (*person*, *fox*, *weight*, *bigger*)) (Forbes and Choi, 2017)

³This is distinguished from the MSE loss used to train regression models, as it is a distance measure over pairs of distributions.

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PS: Can you think of any other estimate that is sufficient enough to extract those details?

		Accuracy		MSE		EMD	
		mcc	rgr	mcc	rgr	mcc	rgr
Lengths	Aggregate	.24	.24	.027	.027	.077	.077
	word2vec	.30	.12	.026	.099	.079	.072
	ELMo	.43	.23	.019	.084	.055	.072
	BERT	.42	.24	.020	.084	.056	.072
	NumBERT	.40	.22	.021	.086	.052	.072
Masses	Aggregate	.15	.15	.026	.026	.076	.076
	word2vec	.26	.20	.025	.088	.082	.077
	ELMo	.36	.21	.021	.087	.061	.077
	BERT	.33	.22	.021	.085	.062	.077
	NumBERT	.32	.20	.021	.088	.057	.077
Prices	Aggregate	.24	.24	.019	.019	.057	.057
	word2vec	.26	.14	.019	.090	.063	.087
	ELMo	.37	.21	.016	.081	.051	.087
	BERT	.33	.19	.017	.083	.054	.087
	NumBERT	.32	.17	.017	.085	.051	.087
Animal Masses	Aggregate	.30	.30	.022	.022	.059	.059
	word2vec	.33	.35	.021	.069	.069	.077
	ELMo	.43	.28	.016	.079	.057	.077
	BERT	.41	.26	.017	.079	.058	.077
	NumBERT	.42	.23	.018	.083	.053	.077

Table 1: Comparison of encoders and probes on the Scalar probing task on DoQ data. Results are averaged over 10-fold cross-validation.

Predictions for this task are made by comparing the point estimates for **rgr** and highest-scoring buckets for **mcc**. The second is an empirical distribution of product price data extracted from the Amazon Review Dataset (Jianmo Ni, 2019). We retrained a model on DoQ prices using 12 power-of-4 buckets to support finer grained predictions.

6 Results

Table 1 shows results of scalar probing on DoQ data. For MSE and EMD the best possible score is 0, while for accuracy we take a loose upper bound to be the performance of a model that samples from the ground-truth distribution and is evaluated against the mode. This method achieves accuracies of 0.570 for lengths, 0.537 for masses, and 0.476 for prices. Compared to the baseline, we can see that **mcc** over the best encoders capture about half (as measured by accuracy) to a third (by MSE and EMD) of the distance to the upper bound, suggesting that while a significant amount of scalar information is available, there is a long way to go to support robust commonsense reasoning.

From Table 1, we see that the more expressive models using **mcc** consistently beat **rgr**, with the latter frequently unable to improve upon the Aggregate baseline. This shows that scale information is present in the embeddings, but training on the median alone is not enough to reliably extract it;

⁴The full set of experimental results are shown in Table 6 in the Appendix.

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the full data distribution is needed.

CV_DRAWING [CV: red] CV_HIGHLIGHT [CV: green] Comparing results by encoders, we see that

CV_DRAWING [CV: red] CV_HIGHLIGHT [CV: green] Word2Vec performs significantly worse than the

CV_DRAWING [CV: red] CV_HIGHLIGHT [CV: orange] contextual encoders – even though the task is non-

CV_DRAWING [CV: red] contextual – indicating that contextual information

during pre-training improves the representation of

scales.

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CV_DRAWING [CV: red] Despite being weaker than BERT on down-

CV_DRAWING [CV: red] stream NLP tasks, ELMo does better on scalar

CV_DRAWING [CV: red] CV_HIGHLIGHT [CV: yellow] probing, consistent with it being better at numer-

CV_DRAWING [CV: red] CV_HIGHLIGHT [CV: yellow] acy (Wallace et al., 2019) due to its character-level

tokenization.

NumBERT does consistently better than ELMo

CV_HIGHLIGHT [CV: yellow] and BERT on the EMD metric, but worse on MSE

and Accuracy. This is in contrast to other standard

benchmarks such as Q/A and NLI, where Num-

CV_HIGHLIGHT [CV: green] BERT made no difference relative to BERT. Our

CV_HIGHLIGHT [CV: green] key take away is that the numerical representation

has an impact on scale prediction (see Figure 2

for qualitative differences), but the direction is sen-

CV_HIGHLIGHT [CV: orange] sitive to the choice of evaluation metric. As dis-

cussed in Section 5, we believe EMD to be the most

robust metric *a priori*, but this finding highlights

the need to still examine the full range of metrics.

CV_HIGHLIGHT [CV: pink] Results on Animal Masses (Table 1) show that

CV_DRAWING [CV: red] CV_HIGHLIGHT [CV: pink] training models only on objects in a narrow domain

CV_DRAWING [CV: red] CV_HIGHLIGHT [CV: pink] can significantly improve scalar prediction, under-

CV_DRAWING [CV: red] CV_HIGHLIGHT [CV: pink] scoring the importance of context. For example,

while “crane” in general can refer to either a bird or

a piece of construction equipment, only the former

is relevant in the animal domain, giving the model

a simpler distribution of masses to predict.

Note that, despite significant differences in the

raw numbers for each scale (mass/length/price), the

relative behavior of encoders, metrics and probes

are the same, indicating that our conclusions are

broadly applicable.

Transfer experiments On the F&C relative com-

parison task (Table 2), **rgr**+NumBERT performed

best, approaching the performance of using DoQ

as an oracle, though short of specialized models

for this task (Yang et al., 2018). Scalar probes

trained with **mcc** perform poorly, possibly because

a finer-grained model of predicted distribution is

not useful for the 3-class comparative task. On

the Amazon price dataset (Table 3) which is a full

distribution prediction task, **mcc**+NumBERT did

best on all three metrics. On both zero-shot trans-

fer tasks, NumBERT was the best encoder on all

configurations of metric/objective, suggesting that

manipulating numeric representations can signifi-

dev test

mcc rgr mcc rgr

word2vec	.40	.73	.38	.74
ELMo	.47	.71	.47	.72
BERT	.48	.71	.49	.71
NumBERT	.51	.77	.54	.76
DoQ (Elazar et. al. 2019)	-	.78	-	.77
Yang et. al. '18	-	.86	-	.87

Table 2: Accuracy on VerbPhysics (Forbes and Choi, 2017).

Accuracy MSE EMD

Aggregate	.04	.04	.02	.02	.06	.06
word2vec	.09	.23	.02	.07	.07	.08
BERT	.14	.25	.02	.07	.06	.08
NumBERT	.18	.27	.02	.07	.05	.08

Table 3: Results on consumer price data (Jianmo Ni, 2019).

cantly improve performance on scalar prediction.

7 Conclusion

From our novel scalar probing experiments, we

find there is a significant amount of scale informa-

tion in object embeddings, but still a sizable gap

to overcome before LMs achieve this prerequisite

of CSR. We conclude that although we observe

some non-trivial signal to extract scale information

from language embedding, the weak signals sug-

gest these models are far from satisfying common

sense scale understanding.

Our analysis points to improvements in model-

ing context and numeracy as directions in which

progress can be made, mediated by the transfer

of scale information from numbers to nouns. The

NumBERT intervention has a measurable impact

on scalar probing results, and transfer experiments

suggest that it is an improvement. For future work

we would like to extend our models to predict

scales for sentences bearing relevant context about

scalar magnitudes, e.g. “I saw a baby elephant”.

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using scientific notation for numbers and Jeremiah

Liu for helpful discussions on statistical distance

measures.

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A Model Hyperparameters

Here we provide the model hyperparameters we use for reproducibility.

A.1 Probing Layer of the Scalar Probing Model

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For the regression model, we use a ridge regression with regularization strength of 1. For the multi-class classification model, we use a linear classifier with a softmax activation function and regularization strength of 0.01.

Task	Metric	BERT Base	NumBERT
CoLA	Dev Acc	.745	.742
MNLI	Dev Matched Acc	.791	.789
	Dev Mismatched Acc	.795	.798
MRPC	Dev Acc	.816	.802
Squad v1	F1	.799	.789
Squad v2	Best F1	.669	.673
STS-B	Dev Pearson's r	.866	.871

Table 4: NumBERT vs BERT-base on a suite of standard NLP benchmarks.

Dataset	Subset	#Data Samples
DoQ	all masses	76,424
	all prices	212,277
	all lengths	244,517
	animal masses	519
	product category prices	1,789
Product Price	-	1,888
F&C Cleaned	train	172
	dev	1,267
	test	1,522

Table 5: Statistics of Datasets/Resources used in our paper

For experiments on the narrow domains with smaller datasets, we first use PCA to reduce embeddings down to 150 dimensions before training the probing model.

A.2 NumBERT

NumBERT is pretrained on the Wikipedia and Books corpora used by the original BERT paper (Devlin et al., 2018). The BERT configuration is the same as BERT-Base (L=12, H=768, A=12, Total Parameters=110M). The language model masking is applied after WordPiece tokenization with a uniform masking rate of 15%. Maximum sequence length (number of tokens) is 128. We train with batch size of 64 sequences for 1,000,000 steps, which is approximately 40 epochs over the 3.3 billion word corpus. All the other hyperparameters and implementation details (optimizer, warm-up steps, etc.) are the same as the original BERT implementation. Table 4 shows a comparison of NumBERT vs a re-implementation of BERT-base with identical settings as above, on a suite of standard NLP benchmarks, and we conclude that the two models reach similar performance on these tasks.

B Data Statistics

Table 5 shows the statistics of 3 datasets/resources we use in this paper. For DoQ, we take the original resource and get each subset by filtering using the corresponding dimensions and/or object types (e.g. all objects, animals, product categories, etc.). Also, only objects with more than 100 values collected in the resource are used. For F&C Cleaned dataset, we use the data and the train/dev/test splits from (Elazar et al., 2019).

C Complete Experimental Results

We model the distributions of those scalar attributes as categorical distributions over 12 categories. We first take the base-10 logarithm of all the values and then round them to the nearest integer (between -2 and 9 for all scales). We treat each integer as a bucket and use the normalized counts in each bucket as the true distribution for that scalar attribute of the object.

To explore the effect of ambiguity, we divide all the data in DoQ for each scale into 2 sets, **Unimodal** where the distribution has one well-defined peak and **Multimodal**, where multiple peaks are present. The number of peaks were identified by a simple hill-climbing algorithm.

As words often have more than one meaning in different contexts or even modifiers, their corresponding distribution from DoQ should reflect the different senses if they appeared enough in the data. When the objects are different enough (e.g. an ice-cream have mainly one meaning and its size doesn't vary much, as opposed to a truck which can be a toy truck, which is very small, or an actual vehicle, which is very big), they may have different modalities. In order to better understand our results, we wish to separate between objects of different modalities to objects with a single modality.

In order to estimate a multi-modal function, we take the bucketed DoQ distribution and smooth it into a probability density function. Then, by finding local maxima over the fitted density function, we estimate a distribution to be multi-modal if we find more than one maximum, otherwise we determine it to be a single-modal distribution.

The complete experiment results including the multimodal experiments are in Table 6.

			Accuracy			MSE			EMD		
			All	Multi	Uni	All	Multi	Uni	All	Multi	Uni
Lengths	mcc	Aggregate	.250	.250	.250	.028	.028	.028	.077	.077	.077
		word2vec	.300	.310	.280	.026	.022	.031	.079	.074	.087
		ELMo	.430	.420	.440	.019	.020	.017	.055	.056	.053
		BERT	.420	.410	.420	.020	.021	.018	.056	.058	.054
		NumBERT	.420	.410	.420	.020	.021	.018	.056	.058	.054
		ELMo	.430	.420	.440	.019	.020	.017	.055	.056	.053
	rgr	Aggregate	.250	.250	.250	.028	.028	.028	.077	.077	.077
		word2vec	.120	.120	.130	.099	.100	.097	.072	.070	.074
		ELMo	.230	.230	.240	.084	.085	.082	.072	.070	.074
		BERT	.240	.230	.240	.084	.085	.081	.072	.070	.074
		NumBERT	.240	.230	.240	.084	.085	.081	.072	.070	.074
		ELMo	.230	.230	.240	.084	.085	.082	.072	.070	.074
Masses	mcc	Aggregate	.330	.330	.330	.021	.021	.021	.062	.062	.062
		word2vec	.260	.260	.260	.025	.026	.023	.082	.083	.080
		ELMo	.360	.360	.360	.021	.021	.019	.061	.062	.059
		BERT	.330	.330	.330	.021	.022	.019	.062	.063	.060
		NumBERT	.330	.330	.330	.021	.022	.019	.062	.063	.060
		ELMo	.360	.360	.360	.021	.021	.019	.061	.062	.059
	rgr	Aggregate	.240	.240	.240	.021	.021	.021	.062	.062	.062
		word2vec	.200	.190	.200	.088	.090	.086	.077	.076	.080
		ELMo	.210	.200	.210	.087	.088	.085	.077	.076	.080
		BERT	.220	.210	.220	.085	.086	.084	.077	.076	.080
		NumBERT	.220	.210	.220	.085	.086	.084	.077	.076	.080
		ELMo	.210	.200	.210	.087	.088	.085	.077	.076	.080
Prices	mcc	Aggregate	.240	.240	.240	.021	.021	.021	.062	.062	.062
		word2vec	.260	.250	.280	.019	.014	.024	.063	.055	.072
		ELMo	.370	.360	.380	.016	.018	.013	.051	.053	.047
		BERT	.330	.320	.330	.017	.019	.014	.054	.055	.051
		NumBERT	.330	.320	.330	.017	.019	.014	.054	.055	.051
		ELMo	.370	.360	.380	.016	.018	.013	.051	.053	.047
	rgr	Aggregate	.240	.240	.240	.021	.021	.021	.062	.062	.062
		word2vec	.140	.130	.150	.090	.093	.085	.087	.084	.092
		ELMo	.210	.210	.220	.081	.083	.078	.087	.084	.092
		BERT	.190	.190	.190	.083	.085	.081	.087	.084	.092
		NumBERT	.190	.190	.190	.083	.085	.081	.087	.084	.092
		ELMo	.210	.210	.220	.081	.083	.078	.087	.084	.092
Animals	mcc	Aggregate	.330	.330	.330	.021	.021	.021	.062	.062	.062
		word2vec	.330	.320	.350	.021	.020	.023	.069	.066	.075
		ELMo	.430	.440	.420	.016	.015	.019	.057	.056	.059
		BERT	.410	.390	.450	.017	.016	.019	.058	.057	.060
		NumBERT	.410	.390	.450	.017	.016	.019	.058	.057	.060
		ELMo	.430	.440	.420	.016	.015	.019	.057	.056	.059
	rgr	Aggregate	.240	.240	.240	.021	.021	.021	.062	.062	.062
		word2vec	.350	.350	.360	.069	.069	.069	.077	.077	.077
		ELMo	.280	.250	.330	.079	.080	.077	.077	.081	.070
		BERT	.260	.260	.240	.079	.076	.085	.077	.081	.070
		NumBERT	.260	.260	.240	.079	.076	.085	.077	.081	.070
		ELMo	.280	.250	.330	.079	.080	.077	.077	.081	.070
Household Product Prices	mcc	Aggregate	.470	.470	.470	.010	.010	.010	.046	.046	.046
		word2vec	.510	.490	.540	.008	.008	.008	.041	.041	.041
		ELMo	.540	.520	.570	.007	.007	.007	.038	.038	.039
		BERT	.570	.560	.580	.007	.007	.007	.038	.038	.039
		NumBERT	.570	.560	.580	.007	.007	.007	.038	.038	.039
		ELMo	.540	.520	.570	.007	.007	.007	.038	.038	.039
	rgr	Aggregate	.470	.470	.470	.010	.010	.010	.046	.046	.046
		word2vec	.450	.430	.480	.056	.058	.055	.092	.094	.090
		ELMo	.420	.400	.460	.058	.059	.057	.092	.094	.090
		BERT	.440	.420	.460	.057	.059	.055	.092	.094	.090
		NumBERT	.440	.420	.460	.057	.059	.055	.092	.094	.090
		ELMo	.420	.400	.460	.058	.059	.057	.092	.094	.090

Table 6: Evaluation on all datasets.